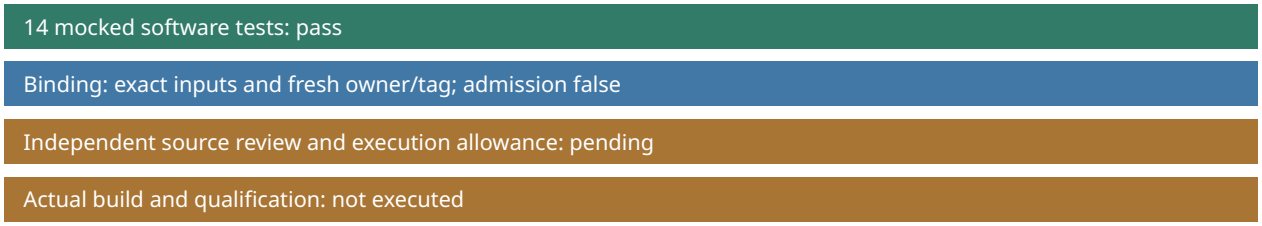
Replacement runtime assembler source checks

Source implementation, 2 November 2026. The runtime-only assembler amendment is implemented with an unadmitted binding to the existing verified context, exact base image/layers, retained source and binary, and a fresh owner/tag. Fourteen mocked software tests pass with unchanged final source hashes. No Docker command, build or image payload was invoked by these tests.



Mocks exercise source decisions; real Docker and subprocess behavior remain untested.

The entry point rejects an unadmitted binding before creating receipts or invoking external tools. Admission joins the exact assembler and input identities to a separately supplied reviewer record, distinct reviewer identity and explicit one-build allowance. The current draft has admitted=false and null review fields. These structural checks do not authenticate a reviewer signature or substitute for independent review.

The workflow checks daemon identity, parent identity/platform/layers and empty OnBuild, then requires the new tag to be absent. It verifies the full staged input readback before the build, before sending the context and after readback. The only build command disables pull and network and uses the pinned base. Returned IID custody, runtime ownership/source labels, configuration and base-layer prefix are checked. A persistent exclusive-create attempt marker prevents reuse of the same owner across new receipt directories.

Tests cover successful one-build flow; rejecting parent build triggers, occupied tags, changed context and wrong output owner; failure and timeout without retry; refusal of unadmitted/duplicate-key inputs; reviewer binding/identity and expiry checks; and one-use marker behavior. Mock tests emulate root-owned IID metadata and Docker replies. They do not test real sudo signals, process reaping, daemon termination or filesystem storage-driver behavior.

The driver records client PID/process group, terminal status and uncertainty. Its proposed bound is 600 seconds including 60 seconds reserved for reconciliation, with at most 240 seconds for the build client and the absolute draft deadline also applied. Output size is monitored; this polling limit is not a hard disk quota. A failed or timed-out build is unresolved; an image-tag readback does not establish daemon/build termination. No automatic retry or image cleanup is implemented.

During source inspection the absolute deadline was added to the driver and preflight checks as well as entry admission. The earlier source and test receipt remain retained; the final fourteen-test result binds the corrected source. Independent source review is still pending. A successful admitted assembly would only produce an image pending qualification; Q1 history and the strict qualification diff rule remain unchanged. The 1,260-target scope and scientific planning limits are preserved.